

PRD_Projects

Projects Module Page

Full Stack Product Requirements Document

Defines the premium project library where users browse every project they are authorized to access. This PRD covers the Projects Module page only; the internal Project Page is governed by PRD_Rework_1.

Confirmed product direction: a clean visual grid, four balanced project cards per desktop row, no local search controls, and a top-right Create New Project action.

Status: Confirmed for implementation
Audience: Product, UI/UX, Frontend, Backend, QA, DevOps
Version: 1.0

1. Document Control

Field	Value
Document	PRD_Projects - Projects Module Page
Primary route	/projects
Entry point	Projects item in the fixed platform sidebar
Primary objective	Let users identify and open authorized projects quickly through a premium image-led grid.
Related PRD	PRD_Rework_1 - Project Page
Out of scope	Detailed Project Page functionality, project creation fields, permissions matrix design, project archiving workflow and map/list views.

2. Product Purpose

The Projects Module is the platform-wide visual library of projects visible to the current user. It is intentionally simpler than a dashboard.

- Make project recognition fast through consistent visual identity.
- Avoid dense metrics, tables and operational widgets on this page.
- Keep project discovery visually premium and easy to scan.
- Open the relevant Project Page with one click.
- Respect company, group and project access rules without exposing inaccessible projects.

The page must feel like a curated portfolio inside an operational platform, not a traditional ERP project list.

3. Confirmed Product Decisions

ID	Topic	Decision
PD-01	View mode	Premium card grid only
PD-02	Desktop density	Four medium balanced cards per row
PD-03	Card content	Project image, project name, location and owning company only
PD-04	Pinned projects	Remain inside the same grid and appear first
PD-05	Local controls	No local search bar, filters or sorting controls
PD-06	Search	Handled by the platform Global Search and its Filters button
PD-07	Create action	Create New Project button in the top-right of the Projects page
PD-08	Create destination	Opens the New Project Setup Flow
PD-09	Card action	Clicking a card opens that project's Project Page

4. Information Architecture and Navigation

Navigation path:

Platform Sidebar -> Projects Module Page -> Project Card -> Project Page

- The Projects item is part of the fixed left navigation.
- The Projects item receives the active-state treatment when the user is on /projects.
- The page inherits the standard platform header, including Global Search, Filters, notifications and user controls.
- The Projects page does not duplicate Global Search or its filtering capabilities.
- Create New Project is a page-level action and remains positioned in the content header, not the global header.

5. Desktop Layout Blueprint

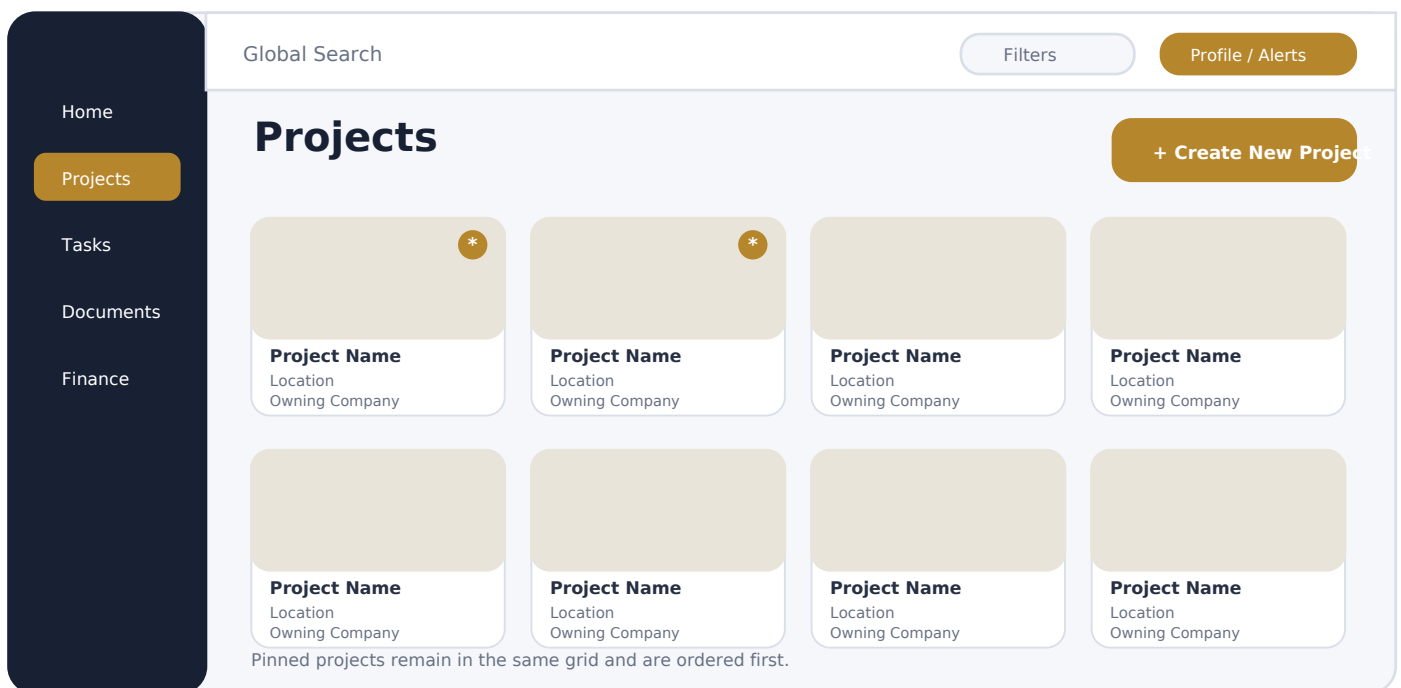


Figure 1. Functional layout blueprint. Visual styling should follow the platform design system and premium Project Page direction.

6. Page Header

- Left: page title Projects.
- Right: primary button Create New Project.
- No subtitle, statistics, filter chips or local toolbar are required for this phase.
- The button must be hidden when the user lacks project.create permission.
- The page header remains visible above the grid and follows the standard responsive shell.

PRJ-PAGE-001 - The page must display the title Projects.

PRJ-PAGE-002 - Create New Project must appear in the top-right on desktop.

PRJ-PAGE-003 - The create action must open the New Project Setup Flow.

PRJ-PAGE-004 - Unauthorized users must not see or access the create action.

7. Project Card Specification

Each card is intentionally minimal. The card must show only the information confirmed below.

Element	Requirement
Project image	Uses the project's pinned thumbnail or configured fallback image. Image crop must be consistent across the grid.
Project name	Primary text. Maximum two lines before ellipsis. Full value available through accessible label or tooltip when truncated.
Location	Secondary text. City/area formatting follows the project record.
Owning company	Displays the legal or configured display name of the company that owns/builds the project.
Pin affordance	A subtle personal favorite control may appear on hover/focus and must not add permanent visual clutter.

Do not add status, progress, budget, task counts, sales data or other metrics to the card in this version.

PRJ-CARD-001 - The entire card must be clickable and keyboard accessible.

PRJ-CARD-002 - Clicking the card must navigate to /projects/{projectId}.

PRJ-CARD-003 - Card hover must use a subtle premium lift, soft shadow and restrained image zoom.

PRJ-CARD-004 - The card must preserve a consistent image aspect ratio and card height.

PRJ-CARD-005 - Missing images must use a branded fallback without breaking the grid.

8. Pinned Project Behavior

- Pins are personal to the authenticated user.
- Pinned projects stay inside the main grid; no separate Pinned Projects section is created.
- Pinned projects are ordered before unpinned projects.
- Pinning or unpinning must update the grid immediately with optimistic UI and rollback on failure.
- Pin state must not affect another user's project ordering.
- Within the pinned and unpinned groups, the system uses a stable backend order.

PRJ-PIN-001 - Pin state must be persisted per user and project.

PRJ-PIN-002 - Pinned projects must appear first on every Projects page load.

PRJ-PIN-003 - The pin action must not navigate into the Project Page.

9. Global Search Integration

The platform Global Search is the only search surface used by the Projects Module.

- Search may return projects together with other platform entities.
- The Global Search Filters button can restrict results by result type, company, project, location, status, department, user and date range where applicable.
- When Global Search is initiated while the Projects Module is active, the interface may prioritize Project as the initial result type without creating a local search control.
- Selecting a project search result opens its Project Page.

Global Search and Global Search Filters remain separate registered platform pages/components and must not be reimplemented inside /projects.

10. Responsive Behavior

Viewport	Grid	Header behavior
Desktop >= 1280px	4 cards per row	Title left; Create New Project right

Viewport	Grid	Header behavior
Laptop 1024-1279px	3 cards per row when required by available content width	Same single-row header where space permits
Tablet 600-1023px	2 cards per row	Compact spacing; action remains visible
Mobile < 600px	1 card per row	Title and action may stack; action remains easy to reach

The confirmed four-card density applies to standard desktop. Lower card counts are responsive adaptations, not additional view modes.

11. Loading, Empty and Error States

State	Expected behavior
Initial loading	Display card skeletons matching the final card geometry. Avoid full-page spinner-only loading.
No accessible projects	Show a calm empty state. Users with create permission receive a Create New Project action; others receive a permissions-aware explanation.
Image loading failure	Use the standard project fallback image without collapsing the card.
Page request failure	Show an inline retry state while retaining the application shell.
Pin mutation failure	Rollback optimistic state and show a non-blocking error notification.

12. Permission and Scope Rules

- Users only receive projects they are authorized to view through company, group, project-membership and explicit access rules.
- The frontend must never rely on visual hiding as the access-control mechanism; backend authorization is mandatory.
- Group-level authorized roles may see projects across owned companies.
- Company-scoped users see only projects available within their company and explicit project assignments.
- Create New Project requires project.create within the applicable company or group scope.
- Opening a stale or unauthorized project link must return an access-aware 403/404 experience without exposing project metadata.

13. Recommended Data Contract

Minimum list item returned by the Projects API:

Field	Type	Purpose
id	UUID/string	Stable project identifier
name	string	Project name
locationLabel	string	Display-ready location
owningCompany	object	Company id and display name
thumbnail	object/null	Pinned image URL, alt text and variants
isPinned	boolean	Current user's personal pin state
displayOrder	number/string	Stable order inside pinned/unpinned groups
permissions	object	At minimum canView and canPin

API-PRJ-001 - GET /api/projects must return only authorized projects and support cursor-based pagination.

API-PRJ-002 - POST /api/projects/{projectId}/pin must persist the current user's pin.

API-PRJ-003 - DELETE /api/projects/{projectId}/pin must remove the current user's pin.

API-PRJ-004 - The list endpoint must order pinned projects first without requiring client-side merging.

14. Performance and Technical Requirements

- Use responsive image variants and modern image formats where supported.
- Lazy-load images outside the initial viewport while reserving their dimensions to prevent layout shift.
- Use cursor-based incremental loading for large project portfolios; do not render the entire dataset at once.
- Keep pin operations independent from full grid reloads.
- Cache safe list data by user scope and invalidate after relevant project changes.
- Maintain stable card positions during image loading and pin mutations.
- Target a responsive interaction experience under normal business network conditions.

Recommended frontend route: /projects. Recommended project route: /projects/{projectId}. Recommended creation route: /projects/new.

15. Accessibility Requirements

- All cards, pin controls and create actions must be reachable by keyboard.
- Visible focus states must match the premium design system.
- Card images require meaningful alt text or an intentional decorative treatment where text already fully identifies the project.
- Truncated names and locations must remain available to assistive technology.
- Color and hover animation must not be the only indicators of interaction.
- Respect reduced-motion preferences by minimizing or disabling card lift and image zoom.

16. Analytics and Audit Events

Event	Purpose
projects_page_viewed	Measure module usage and visible project count
project_card_opened	Track project navigation from the grid
project_pinned	Track personal pin adoption
project_unpinned	Track personal pin removal
create_project_clicked	Track entry into New Project Setup Flow
projects_load_failed	Monitor reliability and retry frequency

17. Acceptance Criteria

- AC-01 - The Projects page uses a card grid as its only view mode.
- AC-02 - A standard desktop viewport displays four medium balanced project cards per row.
- AC-03 - Each card displays only image, project name, location and owning company.
- AC-04 - Pinned projects remain inside the same grid and are shown before unpinned projects.
- AC-05 - Pin state is personal and persistent per user.
- AC-06 - The page has no local search bar, filters, sorting toolbar, list view or map view.
- AC-07 - The platform Global Search and Filters button remain available in the global header.
- AC-08 - Create New Project appears in the top-right for authorized users.
- AC-09 - Create New Project opens the New Project Setup Flow.
- AC-10 - Clicking a project card opens the correct Project Page.

- AC-11 - Unauthorized projects are never returned by the backend.
- AC-12 - Responsive layouts preserve the same card design at lower column counts.
- AC-13 - Skeleton, empty, image-fallback and error states are implemented.
- AC-14 - The grid supports incremental loading without layout instability.

18. QA Test Matrix

Area	Minimum tests
Rendering	4-column desktop; 3-column laptop; 2-column tablet; 1-column mobile; long names; missing images
Navigation	Card click; keyboard Enter/Space; back navigation; deep link to Project Page
Pins	Pin; unpin; reload persistence; separate users; mutation failure rollback
Permissions	Authorized project sets; group scope; company scope; create button visibility; direct API denial
States	Initial skeleton; empty; API failure; retry; broken thumbnail; partial incremental load
Accessibility	Keyboard order; focus visibility; screen-reader labels; reduced motion; contrast
Performance	Large project count; image lazy loading; no layout shift; cursor loading

19. Page and Flow Registry

The following names must remain registered for future navigation, permission and implementation planning:

- Projects Module Page - this PRD; route /projects.
- Create New Project - top-right page action.
- New Project Setup Flow - destination of Create New Project; separate future specification.
- Project Page - opened from a card; defined in PRD_Rework_1.
- Global Search - shared platform search surface.
- Global Search Filters - shared filters interface used instead of local Projects filters.
- Project Render Gallery - source of the pinned project thumbnail; defined with the Project Page.

These supporting names are part of the product registry even where their detailed functionality is specified in another PRD.

20. Deferred Features

- List view
- Map view
- Local Projects search
- Local Projects filters and sorting controls
- Custom card fields
- Status and progress on cards
- Separate Pinned Projects section
- Bulk actions
- Project archive management
- Detailed New Project Setup Flow fields

Deferred features may be evaluated later but must not be added to the first implementation of this page without an approved PRD update.

21. Definition of Done

- Frontend behavior matches all confirmed decisions and responsive requirements.
- Backend returns only authorized project cards and supports stable pinned-first ordering.
- Personal pins persist correctly across sessions and devices.
- Create New Project permission is enforced in both UI and backend flow entry.
- Global Search integration is retained with no duplicate local search UI.
- Accessibility, error handling, loading states and analytics events are complete.
- QA acceptance criteria pass on supported desktop, tablet and mobile browsers.
- Page and Flow Registry names are included in routing and product documentation.